EE263 Homework 3

Summer 2026

1. Orthogonal complement of a subspace. If $\mathcal{V}$ is a subspace of $\mathbb{R}^n$ we define $\mathcal{V}^\perp$ as the set of vectors orthogonal to every element in $\mathcal{V}$, *i.e.*,

$$\mathcal{V}^\perp = \{\ x \mid \langle x, y \rangle = 0,\ \forall y \in \mathcal{V}\ \}.$$

a) Verify that $\mathcal{V}^\perp$ is a subspace of $\mathbb{R}^n$.

b) Suppose $\mathcal{V}$ is described as the span of some vectors $v_1, v_2, \ldots, v_r$. Express $\mathcal{V}$ and $\mathcal{V}^\perp$ in terms of the matrix $V = \begin{bmatrix} v_1 & v_2 & \cdots & v_r \end{bmatrix} \in \mathbb{R}^{n \times r}$ using common terms (range, nullspace, transpose, etc.)

c) Show that every $x \in \mathbb{R}^n$ can be expressed uniquely as $x = v + v^\perp$ where $v \in \mathcal{V}$, $v^\perp \in \mathcal{V}^\perp$. *Hint:* let v be the projection of x on $\mathcal{V}$.

d) Show that $\dim \mathcal{V}^\perp + \dim \mathcal{V} = n$.

e) Show that $\mathcal{V} \subseteq \mathcal{U}$ implies $\mathcal{U}^\perp \subseteq \mathcal{V}^\perp$.

2. Coin collector robot. Consider a robot with unit mass which can move in a frictionless two dimensional plane. The robot has a constant unit speed in the y direction (towards north), and it is designed such that we can only apply force in the x direction. We will apply a force at time t given by f_j for $2j - 2 \le t < 2j$ where $j = 1, \ldots, n$, so that the applied force is constant over time intervals of length 2. The robot is at the origin at time $t = 0$ with zero velocity in the x direction.

There are $2n$ coins in the plane and the goal is to design a sequence of input forces for the robot to collect the maximum possible number of coins. The robot is designed such that it can collect the ith coin only if it exactly passes through the location of the coin (x_i, y_i). In this problem, we assume that $y_i = i$.

a) Find the coordinates of the robot at time t, where t is a positive integer. Your answer should be a function of t and the vector of input forces $f \in \mathbf{R}^n$.

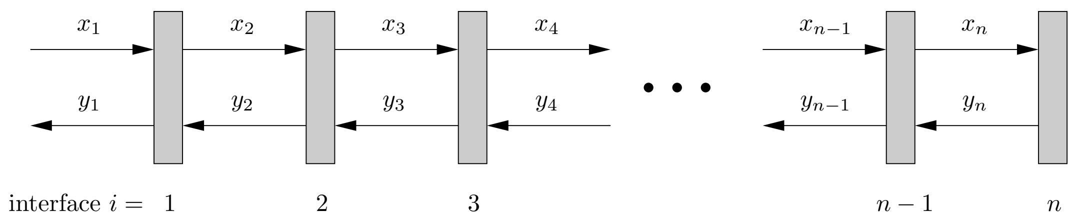
b) Given a sequence of k coins $(x_1, y_1), \ldots, (x_{2n}, y_{2n})$, describe a method to find whether the robot can collect them.

c) For the data provided in `robot_coin_collector.json`, show that the robot cannot collect all the coins.

d) Suppose that there is an arrangement of the coins such that it is not possible for the robot to collect all the coins. Suggest a way to check if the robot can collect all but one of the coins.

e) Run your method on data in `robot_coin_collector.json` and report which coin cannot be collected. Report the input that results in collecting $2n - 1$ coins. Plot the location of the coins and the location of the robot at integer times.

3. **Layered medium.** In this problem we consider a generic model for (incoherent) transmission in a layered medium. The medium is modeled as a set of n layers, separated by n dividing interfaces, shown as shaded rectangles in the figure below.



We let $x_i \in \mathbb{R}$ denote the right-traveling wave amplitude in layer i, and we let $y_i \in \mathbb{R}$ denote the left-traveling wave amplitude in layer i, for $i = 1, \ldots, n$. The right-traveling wave in the first layer is called the incident wave, and the left-traveling wave in the first layer is called the reflected wave. The scattering coefficient for the medium is defined as the ratio $S = y_1/x_1$ (assuming $x_1 \neq 0$).

The right- and left-traveling waves on each side of an interface are related by transmission and reflection. The right-traveling wave of amplitude x_i contributes the amplitude $t_i x_i$ to x_{i+1}, where $t_i \in [0, 1]$ is the transmission coefficient of the ith interface. It also contributes the amplitude $r_i x_i$ to y_i, the left-traveling wave, where $r_i \in [0, 1]$ is the reflection coefficient of the ith interface. We will assume that the interfaces are symmetric, so the left-traveling wave with amplitude y_{i+1} contributes the wave amplitude $t_i y_{i+1}$ to y_i (via transmission) and wave amplitude $r_i y_{i+1}$ to x_{i+1} (via reflection). Thus we have

$$x_{i+1} = t_i x_i + r_i y_{i+1}, \quad y_i = r_i x_i + t_i y_{i+1}, \quad i = 1, 2, \ldots, n-1.$$

We model the last interface as totally reflective, which means that $y_n = x_n$.

a) Explain how to find the scattering coefficient S, given the transmission and reflection coefficients for the first $n-1$ layers.

b) Carry out your method for a medium with $n = 20$ layers, and $t_i = 0.96$, $r_i = 0.02$ for $i = 1, \ldots, n-1$. Plot the left- and right-traveling wave amplitudes x_i, y_i versus i, and report the value of S you find.

   Hint: You may find the matlab function `diag(x,k)` useful.

c) *Fault location.* A fault in interface k results in a reversal: $t_k = 0.02$, $r_k = 0.96$, with all other interfaces having their nominal values $t_i = 0.96$, $r_i = 0.02$. You measure the scattering coefficient $S = S^{\text{fault}}$ with the fault (but you don't have access to the left- or right-traveling waves with the faulted interface). Explain how to find which interface is faulted. Carry out your method with $S^{\text{fault}} = 0.70$. You may assume that the last (fully reflective) interface is not faulty. Be sure to give the value of k that is most consistent with the measurement.

4. **Last year's midterm.** As preparation for this year's midterm, work through last year's midterm, which you can find at `https://ee263.stanford.edu/hw/midterm25.pdf`. This year's midterm will be similar in style: it consists entirely of multiple-choice questions. Last year's exam was designed as a 75-minute exam, and was intended to be of moderate difficulty, challenging in places but not excessively hard.

   For this homework, submit an answer to every question along with a short justification. This does not need to be a formal proof: a sentence or two explaining your reasoning, or a small counterexample, is enough. We want to see the thought process behind each answer, not just the answer itself.